Structured Matrix Nearness Problems : Theory and Algorithms

Nearness

$\min \|A - X\| \quad \text{s.t. } X \in S \qquad X^{\#} = \mathrm{Proj}_S(A)$

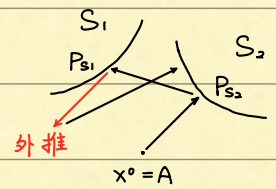
S convex closed, $X^{\#}$ exist and unique

$\min \|A - X\| \quad \text{s.t. } X \in S_1 \cap S_2$

AP (alternative projection)

$X^{k+1} = P_{S_2}(P_{S_1}(X^k))$ 会振荡 (?)

Dykstra S_1, S_2 convex 必收敛

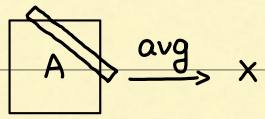


$\min \|A - X\| \quad \text{s.t. } X \in \mathcal{L}$

Linear Structure $\mathcal{L} = \{ X = \sum_{i=1}^{s} x_i u_i \}$ U_i basis

e.g. Toep $U_1 = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} \cdots U_s = \begin{bmatrix} & & 1 \\ & \iddots & \\ 1 & & \end{bmatrix}$

$X^{\#} = \mathrm{Proj}_{\mathcal{L}}(A)$ e.g. $\mathrm{Proj}_{\mathcal{T}}(A)$



Low Rank Problem

$\min \|A - X\| \quad \text{s.t. } X \in \mathcal{L} \cap R_r \qquad R_r = \{ X_{n\times p},\ \mathrm{rank}(X) \le r \}$

Algo 1 : Lift and Proj

就是 AP, $X^{k+1} = \mathrm{trunc_SVD}_r(\mathrm{Proj}_{\mathcal{T}}(X^k))$ 但 R_r 不凸

用 Dykstra 改进, 即 generalized nearness

Algo 2 : Total Least Norm

Algo 3 : Geometric Optimation $\qquad \|\underset{n\times p}{X}\|_{\underset{np\times np}{Q}}^2 = \mathrm{vec}(X)^T Q\, \mathrm{vec}(X)$

$\min_{X_{n\times p}} \|A - X\|_Q \quad \text{s.t. } X \in \mathcal{L} \cap R_r$

$\Updownarrow$ [93] $\qquad f(N)$ only depend on $\mathrm{span}(N)$,

$\min_{\underset{p\times(p-r)}{[N]}} f([N]) \quad \text{s.t. } [N] \in Gr(p, p-r)$ $\qquad$ so denote $f([N])$

(5.8)

$$\bar{f}(N) = f([N]) = \min_X \|A - X\|_Q \qquad (5.7)$$

$$\text{s.t. } X \in \mathcal{L},\ \underset{p\times(p-r)}{XN} = 0$$

$$\mathrm{vec}(XN) = (N^T \otimes I_n)\,\mathrm{vec}(X)$$

$$= (N^T \otimes I_n)\,Ux = \hat{U}x = 0 \qquad (5.9)$$

$\bar{f}(N)$ 有闭式解 $\bar{f}^* = \|A - x^*\|_Q = a^T \hat{U}^T (\hat{U}\hat{Q}^{-1}\hat{U}^T)^{-1} \hat{U} a$

Summary: 求解 $\min_X \|A - X\|_Q$ s.t. $X \in \mathcal{L} \cap R_r$

由 $\mathcal{L}$ 分解为 $U_1 \cdots U_s$, $U_{np\times s} = [\underset{n\times p}{\mathrm{vec}(U_1)} \cdots \mathrm{vec}(U_s)]$

求解 $\underset{np\times 1}{\mathrm{vec}(A)} = U_{np\times s}\, a_{s\times 1}$ 得到 $a_{s\times 1}$

$\hat{Q}_{s\times s} = U^T Q U$

用 Grassmannian Optimization

$$\min_{\underset{p\times(p-r)}{[N]}} a^T \hat{U}^T (\hat{U}\hat{Q}^{-1}\hat{U}^T)^{-1} \hat{U} a$$

$$\text{s.t. } [N] \in Gr(p, p-r),\quad \hat{U}_{n(p-r)\times s} = \underset{p\times(p-r)}{(N^T \otimes I_n)} \cdot \underset{n(p-r)\times np}{} U_{np\times s}$$

得到 $[N^*]$, 进而得到 $\hat{U}^*$

$x^* = [I_s - \hat{Q}^{-1}\hat{U}^T(\hat{U}\hat{Q}^{-1}\hat{U}^T)^{-1}\hat{U}]\, a$

求解 $\mathrm{vec}(X^*) = Ux^*$ 得到 X^*

只能做到 $r = n-1$, 若 $r < n-1$ 就得到 $x^* = 0$

e.g. $\min \|A - X\|$ s.t. $X \in \mathcal{H}(2)$, $\mathrm{rank}(X) \le 1$

$n = p = 2$, $r = 1$, $s = 2$

Hankel 的基 $U_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $\quad U_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$U_{4\times 2} = [\mathrm{vec}(U_1)\ \ \mathrm{vec}(U_2)] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\hat{U}_{2\times 2} = (\underset{1\times 2}{N^T} \otimes \underset{2\times 2}{I_2})U = \begin{bmatrix} n_1 & 0 & n_2 & 0 \\ 0 & n_1 & 0 & n_2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} n_2 & n_1 \\ n_1 & n_2 \end{bmatrix}$$

由 A 确定 a:

$$Ua = \mathrm{vec}(A) \Rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} A_{11} \\ A_{12} \\ A_{21} \\ A_{22} \end{bmatrix}$$

故要求 $A \in \mathcal{H}(2)$

$a = A$ 的 1st 行, 再倒序

$$\min_{[N]} a^T \hat{U}^T (\hat{U}\hat{Q}^{-1}\hat{U}^T)^{-1} \hat{U} a \Rightarrow \min_{n_1, n_2} (a_1^2 + a_2^2)(n_1^2 + n_2^2) + 4a_1a_2n_1n_2$$

$$\text{s.t. } [N] \in Gr(p, p-r) \qquad \text{s.t. } n_1^2 + n_2^2 = 1$$

$\Rightarrow [N] = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ or $\frac{1}{\sqrt{2}}\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ $\Rightarrow$ Only trivial solution $X = 0$

Remedy 故 [121] 针对 $\mathcal{L}$ 为 Hankel 空间

Grassmannian Optimization 步改为

$$\min_{y} \; a^T \tilde{U}^T (\tilde{U}\hat{Q}^{-1}\tilde{U}^T)^{-1}\tilde{U} a \qquad (5.13)$$

$$\text{s.t. } y \in St(r+1, 1) \qquad \underset{(n+p-r-1)\times(n+p-1)}{\tilde{U}} = \begin{vmatrix} y_1 & y_2 & \cdots & y_{r+1} & & \\ & \ddots & \ddots & & \ddots & \\ & & y_1 & y_2 & \cdots & y_{r+1} \end{vmatrix} \qquad (5.14)$$

得到 y^*，进而得到 $\tilde{U}^*$

e.g. $\min \|A - X\|$ s.t. $X \in \mathcal{H}(2)$, $\text{rank}(X) \le 1$

Modified as $\hat{U}_{2\times 3} = \begin{vmatrix} y_1 & y_2 & 0 \\ 0 & y_1 & y_2 \end{vmatrix}$ $\quad \hat{Q} = \hat{U}^T\hat{U} = \begin{vmatrix} y_1^2 & y_1y_2 & 0 \\ y_1y_2 & y_1^2+y_2^2 & y_1y_2 \\ 0 & y_1y_2 & y_2^2 \end{vmatrix}$

$$\min_{y_1, y_2} \; a^T \hat{U}(\hat{U}\hat{Q}^{-1}\hat{U}^T)^{-1}\hat{U} a \qquad \text{s.t. } y_1^2 + y_2^2 = 1$$

$\Rightarrow y^* \Rightarrow \hat{U}^*$ 非线性 LS 问题，用 MATLAB 的 lsqnonlinear 解

Sec 5.3

$$F(N)_{s\times s} = \hat{U}^T\hat{U}_{n(p-r)\times s} \qquad \hat{U}_{n(p-r)\times s} = (\underset{p\times(p-r)}{N^T} \otimes I_n) \underset{n(p-r)\times np}{\cdot} U_{np\times s}$$

$$\min_{X_{n\times p}} \|A - X\|_Q \qquad \text{s.t. } X \in \mathcal{L} \cap R_r$$

$\Updownarrow$ [93]

$$\min_{\underset{p\times(p-r)}{[N]}} f([N]) \qquad \text{s.t. } [N] \in Gr(p, p-r) \qquad (5.8)$$

$$\bar{f}(N) = f([N]) = \min_{X} \|A - X\|_Q \qquad (5.7)$$

$\Updownarrow$

$$\text{s.t. } X \in \mathcal{L}, \quad XN = 0$$

$$\min_{[N], x} \; \frac{1}{2}(a-x)^T \underset{s\times s}{\hat{Q}} (a-x)$$

$$\text{s.t. } (N^T \otimes I_n) U x = \hat{U} x = 0$$

$$[N] \in Gr(p, p-r), \quad x \in \mathbb{R}^s \qquad (5.18)$$

Sec 5.3.1

Augmented Lagrangian

$$G_{\lambda,\mu} = \frac{1}{2}(a-x)^T\hat{Q}(a-x) + \frac{\mu}{2}\|(N^T\otimes I_n)Ux\|_2^2 - \underset{n(p-r)\times 1}{\lambda^T}(N^T\otimes I_n)Ux$$

在 ALM 中，关键问题是

$$\min_{[N],x} \quad G_{\lambda,\mu} \qquad (5.21)$$

$$\text{s.t.} \quad [N] \in Gr(p, p-r), \quad x \in \mathbb{R}^s$$

其中给定 [N] 后，可解得

$$x^*_{s\times 1} = (\hat{Q}_{s\times s} + \mu F_{s\times s})^{-1}(\hat{Q}_{s\times s} a_{s\times 1} + \hat{U}^T_{n(p-r)\times s} \lambda_{n(p-r)\times 1})$$

$$F_{s\times s} = U^T_{np\times s} (\underset{p\times(p-r)}{N} \otimes I_n)(N^T \otimes I_n) U_{np\times s}$$

$$\min_{[N]} \quad f_{\mu,\lambda} = -\frac{1}{2} y^T (\hat{Q} + \mu F)^{-1} y$$

$$(N \otimes I_n)\lambda = vec(\Lambda N^T)$$

$$\text{s.t.} \quad [N] \in Gr(p, p-r), \quad y_{s\times 1} = \hat{Q}_{s\times s} a_{s\times 1} + U^T_{np\times s} vec(\Lambda_{n\times(p-r)} N^T_{p\times(p-r)})$$

$$\Lambda_{n\times(p-r)} = vec^{-1}(\lambda)$$

求导 $\nabla f_{\mu,\lambda\, p\times(p-r)} = -X^T_{n\times p}(\Lambda_{n\times(p-r)} - \mu X_{n\times p} N_{p\times(p-r)})$

$$X_{n\times p} = vec^{-1}(U_{np\times s}\, g_{s\times 1})$$

$$g_{s\times 1} = Ky$$

$$K_{s\times s} = (\hat{Q} + \mu F)^{-1}$$